Dr. Anup Kumar Gupta

Email: reachanupgupta@gmail.com

Google Scholar
anupkumargupta.github.io

Research Interests: Multimodal Affective Computing, Remote Physiological Estimation (rPPG), Trustworthy AI for Healthcare

Education

- **Indian Institute of Technology Indore** — *Aug. 2021 - Sep. 2025*
  Doctor of Philosophy, Computer Science and Engineering — *CPI: 10/10*
  Prime Minister Research Fellowship
  Thesis: Exploring Trustworthy Multimodal Learning for Mental and Physiological Health Assessment
- **Indian Institute of Technology Indore** — *Aug. 2019 - Jul. 2021*
  M.S.(Research) - Computer Science and Engineering — *CPI: 10/10*
  Thesis: Adversarial Attack on Visual Speech Recognition Models
- **Dr. A.P.J. Abdul Kalam Technical University, Lucknow** — *Aug. 2013 - Jul. 2017*
  Bachelor of Technology - Computer Science and Engineering — *Score: 78.42%*

Publications

- **Anup Kumar Gupta**, Puneet Gupta, Abhinav Dhall. "Hierarchical Cross-Attention Transformer for Non-contact Multimodal Pain Classification using Remote Physiological Signals and Visual Features", *International Conference on Automatic Face and Gesture Recognition (2026)*, **Best Student Paper Award**. 🏆
- Trishna Saikia, **Anup Kumar Gupta**, Puneet Gupta, Pasi Liljeberg. "Exploring PPG-Guided Knowledge Distillation for Contactless Respiration Estimation", *International Conference on Automatic Face and Gesture Recognition (2026).*
- Aditya Dixit, **Anup Kumar Gupta,** Puneet Gupta, et al. "RECREATE: Supervised Contrastive Learning and Inpainting-based Hyperspectral Image Denoising," *ISPRS Journal of Photogrammetry and Remote Sensing (2026).*
- **Anup Kumar Gupta**, Trishna Saikia, Puneet Gupta, Abhinav Dhall. "PainXtract: A Multimodal System for Multiclass Pain Classification Using Physiological Signals", *Companion Proceedings of International Conference on Multimodal Interaction (2025).*
- Vaidehi Agarwal, Trishna Saikia, **Anup Kumar Gupta**, Puneet Gupta. "SHINE: Synergizing Transformers with Contrastive Learning for Thriving rPPG-based SpO2 Estimation", *Expert Systems with Applications (2025).*
- **Anup Kumar Gupta**, Ashutosh Dhamaniya, and Puneet Gupta. "RADIANCE: Reliable and interpretable depression detection from speech using Transformer", *Computers in Biology and Medicine (2024).*
- Trishna Saikia, Lokendra Birla, **Anup Kumar Gupta**, and Puneet Gupta, "HREADAI: Heart rate estimation from face mask videos by consolidating Eulerian and Lagrangian approaches", *IEEE Transactions on Instrumentation and Measurement (2023).*
- Aditya Dixit, **Anup Kumar Gupta,** Puneet Gupta, et al. "UNFOLD: 3D U-Net, 3D CNN and 3D Transformer based Hyperspectral Image Denoising," *IEEE Transactions on Geoscience and Remote Sensing (2023).*
- **Anup Kumar Gupta**, Rupesh Kumar, Lokendra Birla, Puneet Gupta. "RADIANT: Better rPPG estimation using signal embeddings and Transformer", *Winter Conference on Applications of Computer Vision (2023).*
- Lokendra Birla, Sneha Shukla, **Anup Kumar Gupta**, Puneet Gupta. "ALPINE: Improving Remote Heart Rate Estimation using Contrastive Learning", *Winter Conference on Applications of Computer Vision (2023).*
- Sneha Shukla, Lokendra Birla, **Anup Kumar Gupta**, Puneet Gupta. "Trustworthy Medical Image Segmentation with improved performance for in-distribution samples", *Neural Networks (2023).*
- Sneha Shukla, **Anup Kumar Gupta**, Puneet Gupta. "Exploring the feasibility of adversarial attacks on medical image segmentation", *Multimedia Tools and Applications (2023).*
- **Anup Kumar Gupta**, Aryan Rastogi, Vardhan Paliwal, Fyse Nassar, Puneet Gupta. "D-NEXUS: Defending text networks using summarization", *Electronic Commerce Research and Applications (2022).*
- **Anup Kumar Gupta**, Vardhan Paliwal, Aryan Rastogi, Puneet Gupta. "TRIESTE: Translation based defense for text classifiers", *Journal of Ambient Intelligence and Humanized Computing (2022).*
- Saumya Mishra, **Anup Kumar Gupta**, and Puneet Gupta. "DARE: Deceiving Audio–visual speech Recognition model." *Knowledge-Based Systems (2021).*

- **Anup Kumar Gupta**, Puneet Gupta, and Esa Rahtu. "FATALRead – Fooling visual speech recognition models." *Applied Intelligence (2021).*
- Jainendra Shukla, Puneet Gupta, Aniket Bera, Arka Sarkar, Prakhar Goel, Shubhangi Butta, **Anup Kumar Gupta**, et al. "Contextual Emotion Learning Challenge." In *16th IEEE International Conference on Automatic Face and Gesture Recognition (2021).*

Patent

- Puneet Gupta, Trishna Saikia, **Anup Kumar Gupta**. "System, apparatus, and method for heart rate determination of individuals wearing face mask", Filed under Indian jurisdiction (Ref. No: 202321071568).

Work Experience

- **Indian Institute of Technology Indore** — *Indore, Madhya Pradesh*
  Translational Research Scholar — *Sep. 2025 - Sep. 2026*
  - Working on SEHAT: Smartphone-enabled remote Heart Rate Assessment and Tracking application.

- **Purdue University** — *Lafayette, USA*
  Visiting Research Scholar — *Jun. 2024 - Jul. 2024*
  - Research visit under the Short Term Research Fellowship.
  - Developed context-aware multimodal sentiment analysis models.

- **Shri G S Institute of Technology & Science (SGSITS)** — *Indore, Madhya Pradesh*
  Guest Faculty and Course Co-ordinator — *Oct. 2021 - Aug. 2025*
  - Teaching under the Prime Minister Research Fellowship (PMRF) scheme.
  - **Courses Taught**: Artificial Intelligence; Soft Computing; Artificial Intelligence and Machine Learning

- **Indian Institute of Technology Indore** — *Indore, Madhya Pradesh*
  Lab Instructor — *Dec. 2020*
  - National Project Implementation Unit (NPIU) courses
  - Technical Education Quality Improvement Programme (TEQIP) courses
  - Conducted lab sessions, hands-on training, and doubt-clearing sessions on Machine Learning and Deep Learning.

- **Kwench Global Technologies Pvt. Ltd.** — *Noida, Uttar Pradesh*
  Software Engineer — *Oct. 2017 - Apr. 2018*
  - Developed Workplace bots as part of a team that is a Product Partner of Workplace, an enterprise connectivity platform by Facebook.
  - Contributed to the Reward and Recognition platform used by over *500* companies across *63* countries.
  - Designed and developed rule-based configurations enabling flexible reward system customization without code changes.

- **Novoinvent Software Pvt. Ltd.** — *Noida, Uttar Pradesh*
  Software Engineer — *Jun. 2017 - Sep. 2017*
  - Worked on a U.S. government insurance project for the state of Maryland, "Maryland Health Connection," the state's official health insurance marketplace.

- **Software Development Centre (Software Incubator)** — *Ghaziabad, Uttar Pradesh*
  Student Co-ordinator/Software Engineer — *Aug. 2014 - May 2017*
  - Developed a Biometric Attendance System for the faculty and staff of Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad.
  - Developed and maintained more than 10 portals, websites, and platforms related to AKGEC.
  - Served as Web Team Member, Event Organiser: Tech Trishna'15, the annual national technical festival of AKGEC, which saw a footfall of 1500+ participants.
  - Developed an online registration and slot booking portal in the Blood Donation Camp, 2016 & 2017.

- **My Own Space Research Consultancy** — *Ghaziabad, Uttar Pradesh*
  Student Intern — *Jun. 2016 - Jul. 2016*
  - Designed and developed a web platform to assist users in accessing property reviews and making informed decisions.

- **Sarvodya Ventures** — *Ghaziabad, Uttar Pradesh*
  Student Intern — *Jun. 2015*
  - Contributed to the development of a live portal connecting transporters, truck drivers, customers, and related stakeholders.

Academic Peer-Review Service

- **Conferences**: Conference on Neural Information Processing Systems (NeurIPS) 🎖, Conference on Computer Vision and Pattern Recognition (CVPR) 🎖, European Conference on Computer Vision (ECCV), AAAI Conference on Artificial Intelligence (AAAI), British Machine Vision Conference (BMVC), IEEE International Conference on Automatic Face and Gesture Recognition (FG), IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME).
- **Journals**: IEEE Transactions on Affective Computing (TAFFC), IEEE Transactions on Human-Machine Systems (THMS), IEEE Transactions on Multimedia (TMM), IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), IEEE Transactions on Instrumentation and Measurement (TIM), IEEE Transactions on Consumer Electronics (TCE), IEEE Journal of Biomedical and Health Informatics (JBHI), ACM Transactions on Computing for Healthcare (TOCH), Information Fusion, Knowledge-Based Systems (KBS), Computer Vision and Image Understanding (CVIU), Biomedical Signal Processing and Control (BSPC), Engineering Applications of Artificial Intelligence (EAAI), Neural Computing & Applications, Applied Computing and Informatics, Journal of Imaging.

Honors & Professional Activities

- **Best Student Paper Award**, IEEE International Conference on Automatic Face and Gesture Recognition (FG 2026), Kyoto, Japan, May 2026.
- **Awarded**, IEEE FG 2026 Broadening Participation Travel Grant, IEEE Biometrics Council, May 2026.
- **Top Reviewer**, Conference on Neural Information Processing Systems (NeurIPS 2026), Datasets & Benchmarks Track, Oct. 2026.
- **Outstanding Reviewer, CVPR 2025, 2026** (Top ∼6% reviewers), May 2025, May 2026
- **Invited Keynote Speaker**, National Conference on "Emerging Technologies (AI, ML & DS)", Vikrant Institute of Technology and Management, Indore, Mar. 2026.
- **Mentored** 20+ B.Tech students; 7 M.S. (Research), M. Tech., M.Sc.; and 6 Ph.D. scholars.
- **Best Technology Award**, Indian Institute of Technology Indore, Feb. 2025
- **Best Research Paper Award**, Indian Institute of Technology Indore, Feb. 2024
- Participated and presented a **highlighted** poster at the **PMRF** Annual Symposium 2024, Mar. 2024
- **Mentor** at Initiative for Research & Innovation in Science (IRIS), Jan. 2024
- **Awarded** Short Term Research Fellowship to visit Purdue University, USA Dec. 2023
- **Student Coordinator** of Karyashala Workshop on Synergizing Healthcare Services and Affective Computing using Deep Learning, Aug. 2023
- **Prime Minister Research Fellowship (PMRF) scholar**, IIT Indore, Aug. 2021 - Aug. 2025
- Completed MS (Research) with **Rank 1** and **perfect CGPA**
- **Student Placement Manager**, Training and Placement Cell, IIT Indore, Sep. 2019 - Aug. 2021
- **Member**, Postgraduate Academic Council, IIT Indore, Nov. 2020 - Feb. 2021
- **Instructor**, in **Rashtriya Avishkar Abhiyan**, an initiative by **Ministry of Education**, Aug. 2019
- **Student Co-ordinator**, Software Development Center (R&D center of college), Aug. 2014 - May 2017